

BUSINESS REPORT

Fraud Network Analysis – Bitcoin Transaction Review

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Reference: Elliptic Bitcoin Transaction Dataset – Compliance Risk Assessment

Purpose

This report presents the findings of a network-based review of Bitcoin transactions from the Elliptic dataset. The objective was to identify transactions exhibiting suspicious financial behaviour and provide actionable compliance insights. The analysis focused on transaction relationships, risk rankings, and behavioural patterns commonly associated with money laundering and financial crime.

Three Highest-Risk Transactions

Transaction 38688623 carries the highest risk score in the network at **0.619**. It received funds from **98 separate source transactions** and redistributed those funds to **15 destinations**. This transaction acts as a major collection and redistribution point within the network. Such behaviour is commonly associated with layering activities, where funds are moved through multiple pathways to make tracing their origin more difficult. Although labelled as licit, its network behaviour is highly unusual and warrants further investigation.

Transaction 36364435 recorded the second-highest risk score at **0.515**. It received funds from **97 different sources** and distributed them to **11 destinations**. The transaction occupies a highly influential position within the network and exhibits a similar pattern to Transaction 38688623. The concentration of incoming funds followed by redistribution raises concerns regarding possible attempts to obscure transaction flows.

Transaction 225859042 ranked third with a risk score of **0.450**. Unlike the first two transactions, this node is classified as **unknown**, meaning its legitimacy cannot be confirmed. Despite this uncertainty, it holds one of the highest PageRank values in the network, indicating substantial influence over transaction flow. Its position makes it a high-priority candidate for compliance review.

Most Suspicious Cluster

The most suspicious cluster identified was centred around **Transaction 38688623** and a group of highly connected transactions including **36364435, 38538087, 47066538, and 73227485**. These transactions repeatedly appeared among the highest-risk nodes and layering candidates. Collectively, they function as major collection and redistribution points, receiving funds from numerous sources before forwarding them onward. This concentration of activity suggests coordinated fund movement designed to complicate transaction tracing and increase anonymity.

Recommendations

1. Enhanced Monitoring and Investigation

Transactions **38688623, 36364435, and 225859042** should be prioritised for immediate review. Their high-risk rankings and central positions within the network indicate elevated compliance risk.

Owner: Head of Transaction Monitoring

2. Investigation of High-Risk Unknown Transactions

Conduct a targeted investigation of unknown-class transactions identified through the risk ranking and fan-out analysis, particularly **225859042** and **102570**. These transactions may represent previously unidentified high-risk entities.

Owner: Financial Crime Investigation Team

Data Limitation

A significant limitation of this analysis is that approximately **65% of the analytical subgraph consists of transactions classified as unknown** (3,364 out of 5,207 transactions). As a result, many suspicious patterns are identified based on transaction behaviour and network position rather than confirmed fraud labels. The findings should therefore be treated as strong investigative leads requiring additional verification before regulatory or enforcement action is taken.